

Blockhouse Internship Task 1

Conceptual Questions

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1. Why measure OFI at multiple depth levels of the order book?

Looking only at the best-level OFI (i.e., level 0) gives a limited picture of what's going on in the order book. In reality, there's a lot of trading interest sitting deeper in the book—levels 1 through 10—that can move prices, especially when large orders hit the book or when liquidity is thin at the top.

By measuring OFI at multiple levels, we can capture more of the hidden supply and demand that's not immediately visible at the best bid and ask. The paper by Cont et al. shows that combining these deeper-level OFIs into a single integrated feature using PCA explains price moves much better than just using the top level alone. This is especially useful for building predictive models where deeper liquidity plays a role.

2. Why do the authors use LASSO regression instead of OLS for cross-impact?

In the cross-impact setup, we're looking at the OFIs from many different stocks to see how they affect one stock's return. That means we could have dozens—or even hundreds—of predictors. If we used regular OLS here, we'd run into problems like overfitting or unstable results, especially when there are more variables than observations.

That's where LASSO helps. It not only shrinks the less useful coefficients, but it also drops the ones that don't matter by setting them to zero. This makes the model sparse and easier to interpret—it tells us which few assets actually have meaningful cross-impact. The paper makes a strong case that this kind of sparsity reflects reality better, where only a few assets at any moment are actually moving the price of a given stock.

3. Why is OFI a better short-term return predictor than trade volume?

Trade volume tells you how much was traded, but it doesn't really tell you about the imbalance between buyers and sellers. Two stocks could have the same volume but very different order flow pressures.

OFI, on the other hand, looks at what's happening on both sides of the book: new limit orders, cancellations, price changes, etc. It gives a clearer sense of who's in control—buyers or sellers. This is super important in the short term where price movements are driven by these imbalances.

Cont et al. show that OFI has a tighter link to short-term returns than trade volume. It reacts faster to what's going on in the market and gives a more accurate signal of near-term direction.